DOD-B-24531/1A(SH) METRIC
19 December 1977
SUPERSEDING
DOD-B-24531/1(SH)
23 February 1976

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, SILVER-ZINC ALKALINE (EMERGENCY), FOR DSV:

Ⓐ TYPE 27 V - 12 Ah (METRIC)

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the battery described herein shall consist of this document and the latest issue of DOD-B-24531.

REQUIREMENTS:

Electrical rating: The rating of each 18-cell battery shall be not less than that shown in table I.

TABLE I. Electrical rating.

Rate ampere (A)	Time hours (h)	Capacity ampere-hour (Ah)	Minimum battery voltage (V)	Minimum average cell voltage 1/ (V)	Minimum cell voltage (V)
6	2	12	23.4	1.45	1.30
12	1	12	21.5	1.35	1.20

1/ The "average voltage" is defined as the average of voltage readings taken at 1/2-hour intervals, starting with the first reading taken between 1/2 and 1 minute after applying the load and including the cut-off voltage.

Note: The minimum battery voltage during a 50-A pulse discharge (10 seconds) is 20 volts.

Life: 600 Ah total output or 1 year, whichever occurs first, before the capacity drops to 70 percent of rated capacity.

Dimensions (in millimeters) (mm) (maximum envelope):

Ⓐ Maximum length - 237
Maximum width - 105
Maximum height - 123
(including terminals)

Mass: Maximum 6 kilograms (wet).
Ⓐ Maximum 4.6 kilograms (dry).

Terminals:
Size: 1/4-28 (two nuts per terminal post for each cell are required).
Maximum height: 12 mm above the cell cover.

Ⓐ Zinc electrodes: Mercury shall not be used. A NAVSEC substitute will be specified.

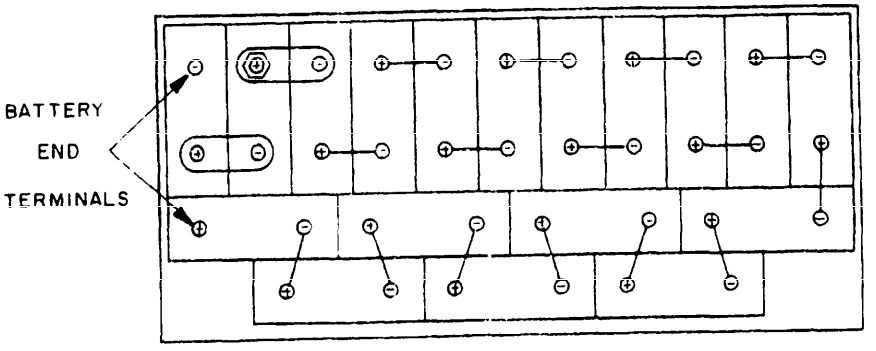
Note: Battery case and cover not required. The cells shall be passed with intercell connectors and banded together to form an 18-cell battery as shown on figure 1, to fit the maximum envelope dimensions.

Ⓐ denotes changes

Preparing activity:
Navy - SH
(Project 6140-N472-1)

FSC 6140

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INTERCELL CONNECTION: TYPICAL ONLY
BATTERY END TERMINAL LOCATION: REQUIRED.
SCALE: NONE.

SH11013

FIGURE 1. Cell and intercell connection layout.

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